Statistics 280 Winter 2026 Midterm 1

Name (Please print legibly)	UMID Number

- Please answer all questions in the space provided and indicate where in the exam you have written additional work.
- **Solutions without justification will be marked as wrong for all questions except 5c. Show all work. A calculator command does not count as justification**.
- Express answers as fractions or as decimals rounded to 2 decimal places.
- The last page is a table of values from the normal distribution.

Academic Integrity

Please read and sign the following statement regarding academic integrity to indicate that you have read, understand, and comply with the statement.

I have never given nor received unauthorized aid on this examination, nor have I concealed any violations of the academic integrity policy as stated by the College of Literature, Science, and the Arts. I have not done anything that may be understood as any action or attempted action that may result in creating an unfair academic advantage for oneself or an unfair academic advantage or disadvantage for any other member of members of the academic community. I understand that if I as suspected of academic misconduct, the incident may be reported to the Office of the Assistant Dean.

Signature: _____

1.(30 points) The Health Insurance Plan of Greater New York ran large-scale trial of screening for breast cancer prevention in 1963. They randomized 62,000 women age 40 to 64 into two equal groups. Women in the treatment group were encouraged to come in for annual screening, including examination by a doctor and X-rays. About 20,200 women in the treatment group did come in for the screening, but 10,800 refused. The control group received usual health care. The study followed the women for many years, with the results for the first 5 years shown in the table below. Rates are per 1,000 women

		Cause of death			
		Breast cancer		All other	
Treatment group		Number	Rate	Number	Rate
Examined	20,200	23	1.1	428	21
Refused	10,800	16	1.5	409	38
Total	31,000	39	1.3	837	27
Control group					
Total	31,000	63	2.0	879	28

Epidemiologists working on the study found that: 1) screening had little impact on other diseases, 2) poorer women were less likely to accept screening than richer ones, and 3) most diseases fall more heavily on the poor than the rich. Please answer the following questions.

a) Does the evidence suggest that screening saves lives? What numbers prove your point?

b) The death rate from all causes among women who accepted screening is about half the death rate among women who refused. Did screening cut the death rate in half? If not, then what explains the difference in death rates?

c) Why would comparing the Examined from the Treatment group to the Refused be a biased comparison? Would it bias in favor of or against screening? Explain.

d) The first year of monitoring found 67 breast cancers in the Examined group, 12 in the Refused group, and 58 in the Control group. Would you conclude that screening causes breast cancer? Explain.

2.(30 points) a) The university is interested in comparing two groups of math and statistics majors. The math majors in group 1 have an average GPA of 3.2 while the math majors in group 2 have an average GPA of 3.0. The statistics majors in group 1 have an average GPA of 3.3 while the statistics majors in group 2 have an average GPA of 3.1. Does it follow that the average GPA in group 1 is higher than the average GPA in group 2? Explain why if so and give a counterexample if not.

2b) Suppose that the average GPA for statistics majors in group 2 is 3.25 and that the other GPAs are the same as in part a). Does it follow that the average GPA in group 1 is higher than the average GPA in group 2? Explain why if so and give a counterexample if not.

2c) (Hypothetical) A college admissions officer at the University of Michigan examines applications from private and public high schools from 2000 to 2020 and finds that a higher percentage of students from private high schools were admitted than from public high schools. She concludes that private high schools provide better preparation for college than public high schools. Is this a controlled experiment or an observational study? Explain.

2d) What is a plausible objection to the argument in part c)?

3. (35 points) a. Use the frequency table below to draw a histogram using the same interval boundaries as in the frequency table. All intervals except the first include the left endpoint but not the right.

Interval	Frequency
0 to 10	6
10 to <30	13
30 to <60	10
60 to <70	13
70 to <100	8

b. Estimate the mean and median for the list of numbers in part a., assuming that the numbers are evenly distributed in each interval.

c. Suppose that you play a game with another person where you guess a number between 0 and 100. The number does not have to be a whole number. You then add the absolute values of the differences between the numbers in the list part a. and your number. The winner of the game is the person with the smaller value for this sum. What number would you guess, assuming that the values in the list are evenly distributed in each interval?

d. Suppose that you play a similar game as in part c., but this time you will double the number you guess and add the squares of the differences from this number. What number would you guess?

4. (35 points) A factory produces medal rods that are supposed to be 50 cm long. Due to small variations in the machining process, the actual lengths of rods are approximately normally distributed with a mean of 50 cm and a standard deviation of 0.2 cm.

a. Estimate the percentage of rods that are between 49.6 cm and 50.4 cm.

b. Estimate the percentage of rods that are between 49.9 cm and 50.2 cm.

c. Estimate the 80th percentile for rod length.

Suppose that the mean income in a country is $60,000 with a standard deviation of $15,000 and that no one has negative income for parts d. and e. **Do not assume that incomes are normally distributed.**

d. What is the maximum possible percentage of citizens with an income of $100,000 and over? What would have to happen for this maximum possible percentage to occur?

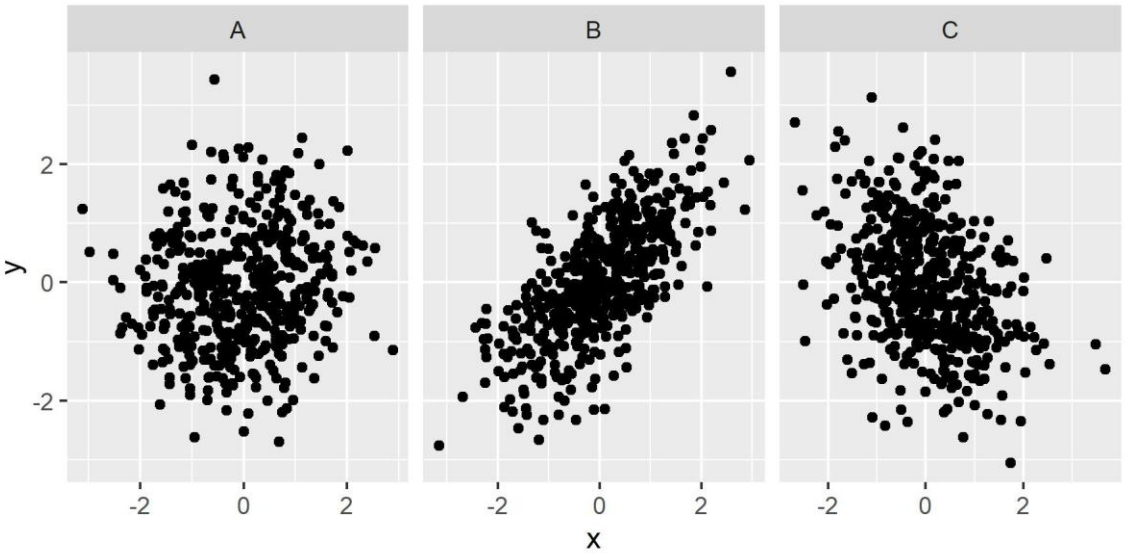
e. What is the maximum possible percentage of citizens with an income under $15,000 or over $95,000?

5. (30 points) a) Find the value of x_2 that will make the correlation between x and y 1 in the table below or explain why there is no such value.

x	y
1	6
x_2	10
3	14
5	26

b) Find the value of x_2 that will make the correlation between x and y 0 in the table below or explain why there is no such value.

c) Match the correlations -0.3, 0.1, and 0.7 to the scatter plots below



6. (40 points) Suppose that the correlation between cups of coffee drank per day and steps walked per day is 0.2 and that the mean and standard deviation for cups of coffee per day are 2 cups and 1 cup, respectively, while the corresponding values for steps walked per day are 6,000 and 1,200. Assume that the scatter plot for cups of coffee and steps walked per day is football shaped.

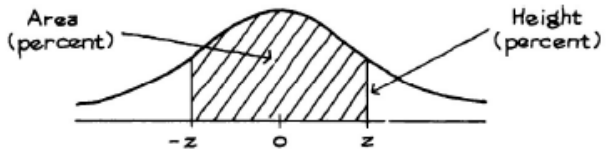
6a) How many steps per day would you predict for someone who drinks 3.2 cups per day?

6b) Give an equation for the predicted relationship between cups of coffee per day and steps walked per day.

6c) Estimate the 10th and 90th percentiles for steps walked per day for someone who drinks 3.2 cups of coffee per day.

6d) Suppose that someone is in the 80th percentile for cups of coffee per day. Estimate the chance that that person walks more than 8,000 steps per day.

Tables



A NORMAL TABLE

z	*Height*	*Area*	*z*	*Height*	*Area*	*z*	*Height*	*Area*
0.00	39.89	0	1.50	12.95	86.64	3.00	0.443	99.730
0.05	39.84	3.99	1.55	12.00	87.89	3.05	0.381	99.771
0.10	39.69	7.97	1.60	11.09	89.04	3.10	0.327	99.806
0.15	39.45	11.92	1.65	10.23	90.11	3.15	0.279	99.837
0.20	39.10	15.85	1.70	9.40	91.09	3.20	0.238	99.863
0.25	38.67	19.74	1.75	8.63	91.99	3.25	0.203	99.885
0.30	38.14	23.58	1.80	7.90	92.81	3.30	0.172	99.903
0.35	37.52	27.37	1.85	7.21	93.57	3.35	0.146	99.919
0.40	36.83	31.08	1.90	6.56	94.26	3.40	0.123	99.933
0.45	36.05	34.73	1.95	5.96	94.88	3.45	0.104	99.944
0.50	35.21	38.29	2.00	5.40	95.45	3.50	0.087	99.953
0.55	34.29	41.77	2.05	4.88	95.96	3.55	0.073	99.961
0.60	33.32	45.15	2.10	4.40	96.43	3.60	0.061	99.968
0.65	32.30	48.43	2.15	3.96	96.84	3.65	0.051	99.974
0.70	31.23	51.61	2.20	3.55	97.22	3.70	0.042	99.978
0.75	30.11	54.67	2.25	3.17	97.56	3.75	0.035	99.982
0.80	28.97	57.63	2.30	2.83	97.86	3.80	0.029	99.986
0.85	27.80	60.47	2.35	2.52	98.12	3.85	0.024	99.988
0.90	26.61	63.19	2.40	2.24	98.36	3.90	0.020	99.990
0.95	25.41	65.79	2.45	1.98	98.57	3.95	0.016	99.992
1.00	24.20	68.27	2.50	1.75	98.76	4.00	0.013	99.9937
1.05	22.99	70.63	2.55	1.54	98.92	4.05	0.011	99.9949
1.10	21.79	72.87	2.60	1.36	99.07	4.10	0.009	99.9959
1.15	20.59	74.99	2.65	1.19	99.20	4.15	0.007	99.9967
1.20	19.42	76.99	2.70	1.04	99.31	4.20	0.006	99.9973
1.25	18.26	78.87	2.75	0.91	99.40	4.25	0.005	99.9979
1.30	17.14	80.64	2.80	0.79	99.49	4.30	0.004	99.9983
1.35	16.04	82.30	2.85	9.69	99.56	4.35	0.003	99.9986
1.40	14.97	83.85	2.90	0.60	99.63	4.40	0.002	99.9989
1.45	13.94	85.29	2.95	0.51	99.68	4.45	0.002	99.9991